AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

class: [16838] germanische sprachen im vergleich
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term: WS25/26

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Preliminary

inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

```
# 20250125(15.03)
### wks.
t
```

Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

```
# t1
load(paste0(Sys.getenv("GIT_TOP"),"/SPUND-LX/germanic/H
```

Methods / Data

our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt (to gemini API) you see in C1.

we first devised model typical lemmata in the AI corpus which are distinct for that corpus using a linear regression model that calculates a score for each lemma in the corpus, see Figure 3 and Figure 2.

```
#params
# lm2<-lmer(f.rel~target*in.gp+(1|size),lmdf.c)
df1<-cbind(df1,"human")
```

C1: System prompt: You are a member of german parliament. Prepare a summary of the text provided to present at a local community meeting of your party members. Output in german language, no preamble, no extra information, just the plain text. Wordcount maximal 300 words, containing not more than 5% of the keywords of the text provided and explicitly not just a list of keywords but an entertaining text. You are supposed to interprete freely, including background insights on daily politics. Keep in mind thatthe text will be used as is as keynotes to the talk being held to the locals. Text:

```
#pc1<-s2$coefficients[1,1]
  geom_density(alpha = 0.3) +
# -H "Content-Type: application/json" \
fj11<-fj6
                    family = poisson(link = "log"),
# tdf4<-getdf(range)
    tdb6$f.norm_within[r]<-mr
#?pivot_wider
tokens.r<-tokens$t1
#fs2$target[match(fa,fg)]<-"gpt"
# range<-c("2021-06-01","2021-07-31")
```

corpus stats

target	freq
human	0.3708
post	0.3890
DIFF:	0.0182

```
word, dfa.pos$token)]
```

p2

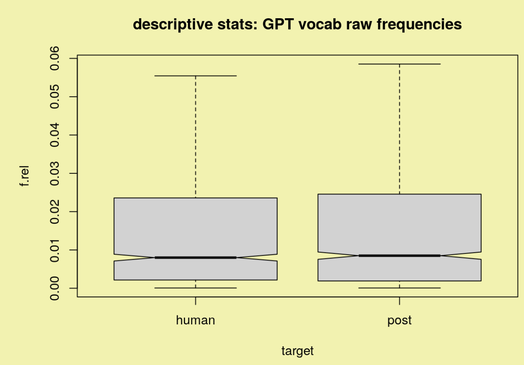

Figure 1: stats descriptive

```
cores
),
zer =
ening/
are valid p-value of freq vs gpt.score, here: p < pt
))
```

p3

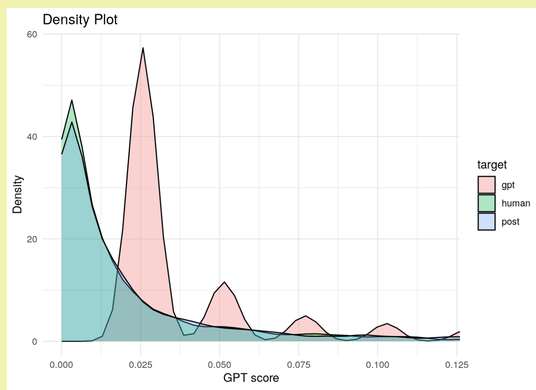

Figure 2: model vocab density

p4

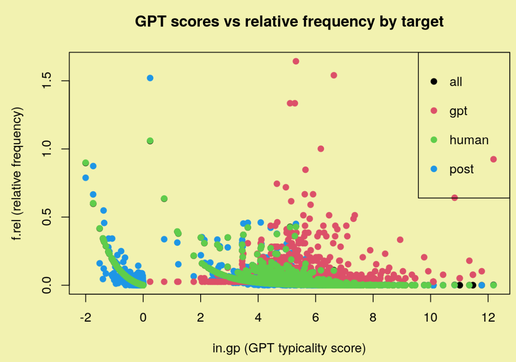

Figure 3: model vocab distribution

Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

{placeholder}{we also found significant frequency increases using a linear regression model, but after experiencing issues with udpipe which first become visible when revisiting our data we had to lemmatize corpus again using the stanza nlp pipeline. results will show up here when evaluated with new lemmas.}

references

BSI, Brand Science Institute. 2025. “Wie KI Unsere Sprache Verändert – Eine Empirische Studie.” https://www.bsi.ag/cases/104-case-studie-wie-ki-unsere-sprache-veraendert---eine-empirische-studie.html.

DIP. 2026. “DIP - Bundestagsprotokolle.” Docs. *DIP - API*. Berlin. https://dip.bundestag.de/%C3%BCber-dip/hilfe/api#content.

Yakura, Hiromu, Ezequiel Lopez-Lopez, Levin Brinkmann, Ignacio Serna, Prateek Gupta, Ivan Soraperra, and Iyad Rahwan. 2025. “Empirical Evidence of Large Language Model’s Influence on Human Spoken Communication.” arXiv. https://doi.org/10.48550/arXiv.2409.01754.